



Advanced Web Development

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2.7.0 Foundations of User Interface Design

Human Interface: 4 Laws

1. Computer shall not harm your work (even with inaction)
2. Computer shall not waste your time or require more work
3. Interface is human if it's responsive to needs (priorities)
4. User should set the pace of interaction

Sampling of Design → Methodologies

<u>Human-centered</u>	<u>User-centered</u>	<u>Usage-centered</u>
human physiological facts	Design Understanding of users (needs, desires, limitations)	User intentions and usage patterns abstraction prototypes

2.7.1 - UI Design

Design Methodologies

It's relevant to avoid ineffective approaches database driven, platform driven designs

User Research Goals

- Understand different user classes and their goals
- Learn their language, skill levels, and attitudes
- Study experiences with previous systems

User Behavioral Patterns

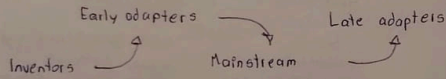
• Safe exploration	• Changes in Midstream	• Habituation
• Instant Gratification	• Deferred Choices	• Spatial Memory
• Satisficing		• Streamlined Repetition
	• Keyboard only	
	• Prospective Memory	

People → Typical user profiles
 Metaphors → Familiar environments
 Interface forms → Recognizable styles

Key Concepts
 → Satisfy goals effectively

2.07.01 - Cognetics and Locus of Attention

GUI, OO concepts and functional programming emerged in 1960's
 Revolutionary ideas require paradigm shifts and go through stages:



Human-Computer Interfaces (HCI)

Key Limitations

MEMORY → Short-term memory holds only 7 things
 Human chunk information for easier recall

PERCEPTION → Locus of attention
 Humans have a single conscious focus
 Shifting attention takes measurable time

COGNITION → Single conscious flow of thought at a time

Design Principles

- Avoid long lists
- Avoid modes (as caps lock)
- Enable automation for routine behaviors (ctrls)
- Provide "in the world" knowledge vs "in the head" memory
- Use familiar terms and user domain terminology
- Design systems allowing users to concentrate on their jobs

2.07.02 - Information Organization

Content Organization Process

- Break scenarios into logical "chunks" (conversational units)
- Look for recurring chunks, keep them loosely coupled on cohesive
- Use metaphors and idioms as starting points
- Avoid mixing metaphors within the same conversational unit

Physical Window Structures

Multiple window One-window paging Tiled panes

Key UI Organizational Patterns

Feature, Search and Browse



Three elements on main page for long lists

New Stream



Reverse chronological for timely content (social apps)

Picture Manager



Thumbnails for collections

Dashboard



Information-dense display from multiple sources

Canvas Plus Palette



For graphical editors



Guide through complicated tasks

Two-panel selector



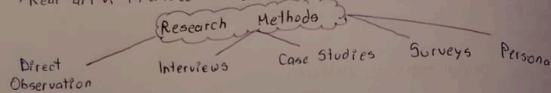
List + details for limited real state

Alternative Views

Let users choose between views of same data

2.07.03 - User Research + User Goals

"Real art of interface design lies in solving the right problem"



User Postures

Sovereign Posture → Expert
 Frequent users

Transient Posture → Occasional
 Beginner users

Critical Design Consideration

- Match vocabulary to user knowledge level
- Don't force sophisticated features on beginners
- Respect low-interest users

2.07.03 - Page Layout

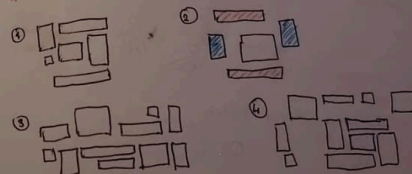
- Visual Hierarchy
 - Visual Flow
 - Grouping and Alignment
 - Dynamic Displays
 - Integration of these elements
- Core Elements

Visual Hierarchy Mechanisms

- Upper-left-corner preference (most important info/items)
- Whitespace, contrasting fonts (bigger / bolder → more imp.)
- Positioning, alignment, indentation
- Graphics (boxes, lines)

Gestalt Principles for Grouping

- PROXIMITY Put related things close together
- SIMILARITY Make similar things look similar
- CONTINUITY Align elements along continuous lines
- VISUAL CLOSURE Humans want to see closed forms



Key Layout Patterns

Visual Framework Consistent layout across pages

Center Stage Main content in largest area

Titled Sections Strong visual titles for sections

Card Stack Stacked panels with tabs

Responsive Disclosure / Enabling Show/enable UI progressively

Liquid Layout Resize with window

Grid of Equals Matrix arrangement with equal visual weight

Navigation

Where now → Where to go → How I get there → Am I There

Signposts

- Window titles, logos, branding devices
- TABS (current highlighted)
- Selection indications

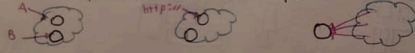
Wayfinding vs Wayshowing

What users do to find their way toward goals

What developers do to facilitate wayfinding

Navigational Models

- ▷ Hub and Spoke, Fully Connected, Multi-level
- ▷ Stepwise, Pyramid, Pan and Zoom
- ▷ Flat Nav, Modal Panel
- ▷ Clear Entry Points, Bookmarks, Escape Hatch



Key Patterns

- Clear Entry Points Few main entry points for 1st time users
- Menu Page List of links as table of contents
- Pyramid Sequential pages viewable in any order
- Modal Panel Force user action before continuing
- Deep-Linked State Capture app state for return later
- Escape Hatch Button to exit from any page
- Sequence Map Show progress in wizards (processes)
- Breadcrumbs Show hierarchical path (top to current)
- Annotated scrollbar scrollbar as nav aid

Actions and Commands

- * Follow GUI conventions when appropriate
- * Make desired actions available and easy to find
- * Label actions well
- * Support sequences, automation and repetition

Core Principles

Action Mechanisms

Visible

- Buttons, menus, pop menu
- Drop-down menu buttons
- Toolbars
- Links, Action Panels

Invisible

- Single/Double clicking
- Keyboard actions
- Drag and Drop
- Typed commands

Basic Guidelines

Follow conventions whenever possible

Use pseudo-3D shading and drop shadows for raised appearance

Change mouse pointer when hovering over clickable items

Use tooltips for descriptive text

Showing Complex Data

Present data visually to impart knowledge using graphics, tables, trees, maps

Use spatial relationships to convey meaning

Organizational Models

LINEAR Lists, single-variable plots

TABULAR Spreadsheets, multi-column lists, sortable tables

HIERARCHICAL Trees, cascaded lists

NETWORK Directed graphs, flow charts

GEOGRAPHIC / SPATIAL Maps, schematics

Preattentive Variables (Visual encoding)

- Color (Hue, Saturation, Brightness)
- Texture, Position and Alignment
- Orientation, Size, Shape

Navigation Techniques

- ▷ Scroll and pan, Zoom
- ▷ Open / Close points of interest
- ▷ Drill down, Magnify

Data Manipulation

SORTING

- Alphabetically
- Numerically
- By Date
- Location
- Category

FILTERING / QUERYING

- Highly interactive
- Iterative
- Contextual

USER - SPECIFIED ARRANGEMENT